

RANBIR KUMAR

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PROFESSIONAL SUMMARY

AI-focused software engineer (B.Tech Computer Science, CGPA 8.28) who takes systems from research to production. Architected and deployed a RAG document-intelligence platform serving real users, a full-stack CMS running an industry body's live website behind a CI/CD pipeline, and a graph-routing navigation app deployed for a national fintech conference; implemented Prototypical Networks from scratch and trained a ResNet-18 encoder end to end for few-shot medical imaging. Filed patent and peer-reviewed research in vision-based perception and explainable AI, with 4 production vision pipelines shipped on NVIDIA Jetson at <40 ms inference.

EDUCATION

B.Tech, Computer Science (AI & ML) — Vellore Institute of Technology, Chennai | CGPA: 8.28/10 | Aug 2023 – May 2027 (expected)

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, Computer Vision, Explainable AI, Database Management & SQL, Probability & Statistics

EXPERIENCE

Technical Integration Lead | [IORA](#), Noida | May 2026 – Present

- Own technical integration for an nRF9151 LTE-M/NB-IoT safety wearable, driving architecture reviews across PCB, firmware (MQTT over TLS, GNSS, flash-buffered telemetry), and power subsystems while coordinating hardware, firmware, and application teams; also architect and ship the company's AI platform (see Projects).

Software Engineering Intern | [IAMAI](#) (Internet and Mobile Association of India), Delhi | May 2026 – Jul 2026

- Engineered an [autonomous browser-testing agent](#) driving Chromium through Playwright to explore the application, execute safe UI actions from DOM and accessibility snapshots, monitor console and network activity, and emit machine-readable defect reports, automating pre-release validation; ran k6 load, stress, and spike testing plus Burp Suite security testing across staging platforms, and owned feature delivery end to end for the two production web platforms detailed in the Projects section.

Computer Vision Engineer (Intern) | [Proeffico Solutions Pvt. Ltd.](#), Noida | May 2025 – Oct 2025 (on-site May–Jul; remote Aug–Oct)

- Built, profiled, and deployed 4 production computer vision pipelines (speed monitoring, pothole detection, fire detection, workplace safety analytics) on NVIDIA Jetson Xavier NX, optimizing to <40 ms inference latency on constrained edge hardware and replacing manual monitoring with automated real-time decisions across client workflows.

Team Lead, Rover | [Technocrats Robotics](#), VIT Chennai | Aug 2025 – Present (member since Mar 2024)

- Lead a cross-functional autonomy team building the rover perception stack — CV-based soil classification, arrow-marker detection, and an end-effector camera pipeline on Raspberry Pi and an i7 mini-PC — owning architecture and code review; All-India Rank 1 in DD Robocon Round 1, 20th worldwide at IRC 2025, and IRC 2026 Finalist among 100+ teams.

PROJECTS

IORA DocQA — Production RAG Document Intelligence Platform | [FastAPI](#) · [Next.js](#) · [pgvector](#) · [DuckDB](#) · [Multi-LLM](#)

github.com/Ranbirkumar26/iora-docqa

- Architected and deployed an end-to-end RAG platform serving real users: an ingestion pipeline handling parsing, chunking, and embedding generation into a Supabase pgvector store, fronted by a FastAPI retrieval service with a hard coverage gate that refuses to answer when retrieval fails to support the question.
- Extended retrieval with a DuckDB text-to-SQL path for uploaded CSV/XLSX, persistent per-user memory, semantic answer caching, and multi-provider LLM failover across Gemini, Groq, and OpenRouter so a single vendor outage degrades gracefully — hardened across six weeks of production iteration behind approval-based signup and admin RBAC.

IAMAI Publishing Platform — Full-Stack CMS & Content Delivery System | [Next.js](#) · [TypeScript](#) · [Supabase](#) · [Postgres](#) · [GitHub Actions](#)

github.com/Ranbirkumar26/IAMAI_CMS_Public

- Engineered the full-stack CMS powering the live public website of the Internet and Mobile Association of India, designing a normalized Postgres schema across eleven content collections with CMS-managed page blocks driving every public page.
- Implemented authentication with password reset, user and permission management, and role-based access control across CMS routes, then modularized the codebase and shipped it through a GitHub Actions production pipeline to Netlify guarded by a scheduled smoke-test monitor that verifies the live deployment.

TerraView — Indoor Venue Navigation & Multi-Floor Routing Engine | [React 19](#) · [Vite 7](#) · [Graph Pathfinding](#) · [Netlify](#) jio-map.netlify.app ·

github.com/Ranbirkumar26/jio-map

- Modelled an entire convention centre as one routing graph over stall, amenity, gate, and lift-connector nodes so shortest-path search resolves journeys that cross hall layouts through numbered gates and change floors through lifts, rendered as guided previous/next segments — deployed for the Global Fintech Fest at Jio World Centre.
- Built exhibitor-name search over 393 mapped stalls and 225 companies behind a pre-deploy validation script enforcing complete stall-to-company mapping, plus an admin authoring mode for gate and routing metadata; ships as an offline-capable static bundle needing no backend, embeddable directly in native Android and iOS WebViews.

RetinaProto — Few-Shot Medical Image Classifier with Grad-CAM Explainability | [PyTorch](#) · [Prototypical Networks](#) · [ResNet-18](#) · [Grad-CAM](#)

github.com/Ranbirkumar26/Explainable-Fewshot

- Implemented Prototypical Networks from scratch — episodic N-way K-shot sampling, class-prototype computation, and Euclidean-distance inference in a learned embedding space — training a ResNet-18 encoder end to end on hospital-sourced retinal images rather than fine-tuning a packaged pipeline, generalizing to new disease classes from very few labels.
- Built the training and evaluation loop with augmentation and seeded episodic evaluation, integrating Grad-CAM attribution maps that expose the retinal regions driving each prediction; basis of a peer-reviewed journal submission under review.

RESEARCH, PATENTS & ACHIEVEMENTS

Patrolling Bot — AI-Powered Autonomous Campus Surveillance Robot | *Patent filed* — YOLO detection (missing ID cards, fire/smoke, overcrowding) and action recognition for fight detection with real-time alerting, deployed on Jetson Xavier NX and Raspberry Pi with LiDAR/ultrasonic navigation (Python, OpenCV, PyTorch, ROS).

Vision-Based Terrain Analysis | *Accepted; publication forthcoming* — real-time multi-model perception fusing YOLOv8 detection with MiDaS monocular depth estimation for terrain classification in autonomous navigation, optimized for embedded edge deployment.

Achievements: Patent Filed · All-India Rank 1, DD Robocon R1 (100/100) · IRC Top 20 Worldwide 2025, Finalist 2026 · SIH 2025 Finalist · Insightify 6.0 Finalist (IIM Lucknow)

SKILLS

Languages & Software Engineering: Python, TypeScript, JavaScript, Java, SQL; Data Structures & Algorithms, Graph Algorithms & Shortest-Path Routing, Object-Oriented Programming, System Design, REST API Design, Authentication & Authorization (RBAC, RLS), Caching, Performance Optimization, Modular Architecture, Code Review

Frontend: React, Next.js (App Router), Vite, Tailwind CSS, shadcn/ui, HTML5, CSS3, Responsive UI, Client State Management

Backend: FastAPI, Flask, Node.js, REST APIs, WebSockets, Server-Sent Events, Microservices, Background Workers (Celery/Redis)

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, CNNs, ResNet, Prototypical Networks, Few-Shot Learning, YOLOv8, MiDaS, OpenCV, Object Detection, Edge Inference (Jetson Xavier NX, Raspberry Pi), Explainable AI (Grad-CAM, SHAP, LIME)

Generative AI: RAG Pipeline Design, Grounded Generation, Semantic Search, Embeddings, Vector Databases (pgvector, FAISS, Qdrant), LangChain, LlamaIndex, HuggingFace Transformers, Semantic Caching, Text-to-SQL

Databases, Cloud & DevOps: PostgreSQL, Supabase (Auth, RLS, Storage), DuckDB, SQLite, Schema Design, Pandas, NumPy, Docker, Git, GitHub Actions (CI/CD), Linux, Vercel, Netlify, Railway, Render, Playwright, Selenium, pytest, k6